

Sky image-based solar forecasting using deep learning with heterogeneous multi-location data: Dataset fusion *versus* transfer learning

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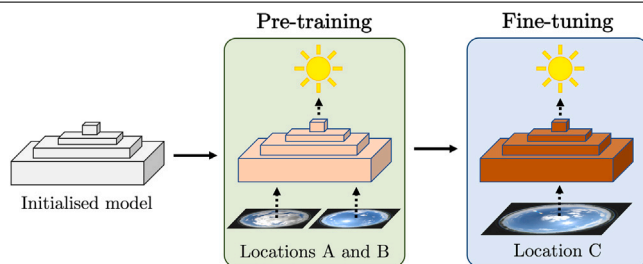
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GRAPHICAL ABSTRACT



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ABSTRACT

Solar forecasting from ground-based sky images has shown great promise in reducing the uncertainty in solar power generation. With more and more sky image datasets available in recent years, the development of accurate and reliable deep learning-based solar forecasting methods using more diverse multi-location data has seen a huge growth in potential. From that perspective, the joint utilization of these heterogeneous data – such as images captured with different camera setups, sensor measurements (i.e., irradiance *versus* photovoltaic power output) varying in scale and distribution – is both a unique opportunity and a critical challenge. This study explores ways to cope with such data heterogeneity and compares three different strategies for training solar forecasting models based on sky image datasets collected across three continents by three research groups. Specifically, for a location of interest, we compare the performance of (1) local models trained individually on a single local dataset (i.e., the standard methodology in the literature); (2) global models jointly trained on the fusion of multiple heterogeneous datasets; and (3) locally fine-tuned models trained via transfer learning from a pre-trained model. The results suggest that, with the current modeling strategy, local models work well when deployed locally, but significant errors are observed when applied offsite. The global model, with proper normalization of the prediction targets, can adapt well to individual locations at the cost of a potential increase in training efforts. Pre-training models on a large and diversified source dataset and transferring to a target location generally achieves superior performance over the other two strategies. With 80% less local

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training data, a fine-tuned model performs similarly to the baseline trained on the entire local dataset. Overall, algorithms built on heterogeneous multi-location sky image datasets have the potential to be more accurate, more robust, and adapt faster to new locations than local models based on a single location.

Abbreviations

ARIMA	Auto-regression integrated moving average
ARMA	Auto-regression moving average
AR	Auto-regression
BatchNorm	Batch normalization
CNN	Convolutional neural network
DEWA	Dubai Electricity and Water Authority
FConv	Freezing Conv. blocks strategy
FC	Fully-connected
FS	Forecasting skill
GHI	Global horizontal irradiance
GT	Ground truth
GW	Gigawatt
MAE	Mean absolute error
MBE	Mean bias error
MSE	Mean squared error
PV	Photovoltaic
ReLU	Rectified linear unit
RGB	Red Green Blue
RMSE	Root mean squared error
RNN	Recurrent neural network
SPM	Smart persistence model
SUNSET	Stanford University Neural network for Solar Electricity Trend
TE	Training effort
TL	Transfer learning
UAE	United Arab Emirates
US	United States
WS	Warm-starting strategy

1. Introduction

The continuous growth of solar photovoltaic (PV) deployment forms a critical part of the global energy transition. According to the International Energy Agency, a record-high 145 gigawatts (GW) capacity has been installed during 2020 even in the face of the pandemic [1]. The global cumulative PV capacity has amounted to 767 GW at the end of 2020, with around 70% of the capacity installed over the last five years [1]. The dramatic rise in PV installations will introduce challenges to the electricity grid due to the intermittency of solar energy, partly caused by clouds passing over solar facilities [2]. To reduce the uncertainty in solar power generation, and further facilitate its integration into the electric grid, accurate and robust solar forecasting is pivotal and urgently needed.

There have been numerous research efforts towards building reliable solar forecasting models since the past decade, aiming at either predicting the future solar irradiance or the power output of PV systems directly. For short-term (i.e. intra-hour) solar forecasting, earlier efforts have used statistical time series methods [3] such as auto-regression (AR), auto-regression moving average (ARMA) and auto-regression integrated moving average (ARIMA) [4], to auto-correlate irradiance/PV measurements for prediction. These methods usually lack forecasting ability as they do not use any information on the movement and distribution of clouds [5]. This limitation can be addressed by analyzing sky images taken from ground-level hemispherical cameras, which offer a panoramic view of the local cloud cover, or infra-red sky cameras [6], which attenuate the effect of the Sun's direct radiation. Early

works proposed to first extract features from sky images, such as red-blue ratio, cloud coverage, cloud motion vectors (e.g., cloud tracking via cross-correlation methods [7] such as particle image velocimetry [8]), and then use these features for building physical deterministic models (e.g., the ladder method [9]). Alternatively, machine learning models have been trained to address this task (e.g., artificial neural networks [10], support vector machine [11], *k*-nearest neighbors [12], XGboost [13]). Furthermore, small networks of all-sky cameras have been used in stereo-vision mode to model the cloud cover in three dimensions, and thus derive local maps of global horizontal irradiance (GHI) [14] or direct normal irradiance [15]. These types of map are validated by shadow camera systems [16].

In the past five years, with the advancements of computer vision techniques, efforts have shifted to building end-to-end deep learning models to directly correlate sky images with target variables (i.e., irradiance or PV power output) [17]. This line of work is featured by using convolutional neural networks (CNNs) to automatically extract patterns from sky images (or sky image sequences) without handcrafted feature engineering. A sizable body of research has been conducted to address various solar modeling tasks, including PV output estimation (i.e., mapping sky images to concurrent PV output) [18], intra-hour PV output [19] and irradiance forecasting [20], covering a wide spectrum of methods, including fusion of imagery and auxiliary data [21, 22], weather-specific models [23], re-sampling and data augmentation strategies [2] and 3D-CNNs [24]. Recent efforts aim at improving cloud dynamics modeling and probabilistic forecast to handle the uncertainty in irradiance [25] or PV output prediction [26]. Studies also investigate hybridizing CNNs with recurrent neural networks (RNNs) such as long short-term memory (LSTM) networks [27], ConvLSTMs [28], spatial gated recurrent units [25,29] to enhance temporal dynamics modeling. More recently, there has been an increasing trend of replacing CNNs and RNNs by transformers to ameliorate the analysis of sky images [30], optical flow [31], auxiliary data [32], and long sequences [33].

Despite the superior performance shown by deep learning-based methods over traditional methods, one of the key limitations is the data [34]. To build a general deep learning model able to perform well not only at the location of the model development dataset but also for unseen data, the training dataset needs to contain a large quantity of cloud cover observations and diversified samples with numerous sky conditions. Contrary to simulated datasets which can be easily extended, real-world data such as sky images are constrained by the period of collection. For this reason, researchers in the solar forecasting community have explored various data augmentation techniques, a common strategy in deep learning, to artificially increase the diversity of image samples via texture changes [2] or geometric transformations [35]. Although data augmentation is viable without accessing additional data, the benefit it could bring is constrained by the diversity of cloud patterns in the dataset. In recent years, efforts from various parties have contributed to the increasing accessibility of solar forecasting datasets, including a virtual testbed [36], tools to facilitate data access such as SolarData [37] and OpenSolar [38], open-source sky image datasets from diverse locations [e.g., California [39,40], New Mexico [41] and Colorado [42] in the United States (US), Switzerland [43] and France [44] in Europe] as well as a comprehensive sky image datasets survey [34]. These potential data sources open new research directions to the academic community such as training a model on a combination of diverse datasets and knowledge sharing via model pre-training and fine-tuning based on different datasets. This opportunity could bridge the gap between academia and solar forecasting or energy companies, which typically operate several sites,

and thus has access to multi-location datasets to train their models on. Better leveraging this type of multi-site datasets therefore holds promise for building more accurate and more robust algorithms while addressing an industrial challenge largely unexplored in an academic context to this day. This paves the way to building models on the equivalent of decades or even centuries of cloud cover observations (compared to a few years of training data at best in current studies).

The primary objective of this work is therefore to explore deep learning approaches based on multi-location datasets in the context of solar forecasting with sky images. Highlighting the advantages and limitations of these methods will facilitate the adoption of multi-location sky image datasets for computer vision (e.g., solar forecasting, cloud masking, cloud motion prediction, radiative transfer modeling). Specifically, the following fundamental questions are examined in this study by using datasets collected globally in three different locations with disparate climate patterns (A description of these three datasets can be found in Section 3):

1. Can trained deep learning models be applied directly to new sites without local re-training or fine-tuning?
2. Should deep learning models be trained locally using the location-specific dataset only or globally via fusion of datasets from different locations?
3. How can global models deal with the dataset heterogeneity, especially the different scales and distributions of prediction targets?
4. Can knowledge be shared between different locations via transfer learning from pre-trained models?

To address these questions, two research groups from Stanford University and Cambridge University, as well as researchers from the Dubai Electricity and Water Authority (DEWA), have worked in parallel and developed deep learning models for a short-term solar forecasting task, the goal of which is to predict solar power (PV power outputs or solar irradiance values) 15-min-ahead based on the sky imagery and measurement data collected in the past 15 min (see Section 4 for more details on the learning setup). Each University team used a different neural network to avoid biases caused by particular model architectures and ensure the reliability of the results reported in this study. The outcomes of this study could provide guidelines for both academia and industry in effectively utilizing multi-location data for training solar forecasting models. Meanwhile, via extensive experiments, this study sheds light on future research towards building a foundation model for solar forecasting, which is pre-trained on a large-scale global dataset and has the potential to be easily adapted to local sites with fine-tuning.

The rest of this paper is organized as follows: in Section 2, the methods for training deep learning models with multi-location data are reviewed, including dataset fusion and transfer learning. Section 3 describes the sky images and PV output/irradiance datasets used in this study from three different locations around the world. Section 4 presents the methodology, including the model architectures, training details and evaluation metrics. Section 5 delves into the experimental designs for exploring the optimal strategies for training PV/irradiance prediction models with multi-location data. Section 6 analyzes and discusses the results and provides directions for future research. Finally, the findings of this study are summarized in Section 7.

2. Review of multi-location dataset modeling

Two methods for multi-location dataset modeling are reviewed in this section. The first type is dataset fusion, basically integrating the datasets from multiple locations and training the model jointly. This approach is based on the expectation that fused data are more informative than any individual dataset. Another method which is widely used in the deep learning community but not yet well studied in image-based solar forecasting, is transfer learning. In transfer learning, the datasets are trained on sequentially, with the model parameters being passed on from one dataset training to another.

2.1. Dataset fusion

Dataset fusion is commonly used in image-based solar forecasting studies with multi-location datasets. Pothineni et al. [45] experimented with two sky image datasets collected in two regions with identical camera setups: one in Italy and one in the Swiss mountains. Irradiance measurements were used to determine the associated sky conditions to be either clear or occluded by thresholding them with the irradiance values derived from a clear sky model. The authors compared training CNN models on the individual datasets with training on the fused dataset for predicting the 5-min ahead sky conditions, with results showing superior performance from training the model jointly on both datasets. Bansal et al. [46] developed a CNN-LSTM auto-regressive model to predict the satellite spectral channel values at the target sites based on a sequence of past satellite observations. They combined satellite data across 25 solar sites in the US to train one global model and test it on individual locations. The authors suggested that while training location-specific models could identify unique local attributes to improve accuracy, their deep learning model is quite data hungry. In the situation where there is not enough data to train and test on any single location, a global model might be beneficial as it can be applied to any location without re-training.

One point to note here is that most of the existing studies using dataset fusion methods are based on the premise that the prediction targets are of similar scales, e.g., solar irradiance or satellite spectral channels for the studies mentioned in the previous paragraph. Very few projects have studied fusing datasets associated with diverse target variables in terms of scale and physical quantities, e.g., PV output measurements from systems with different capacities, or PV output and irradiance measurements. In addition, the imagery data collected by these studies are generally based on similar camera setups, e.g., camera model and placement orientation. However, future studies might fuse multiple data streams generated by different camera setups (e.g., resolution, contrast, color balance, spectral range, orientation). Developing methods able to handle heterogeneous sky camera datasets is therefore essential for the field. If the tasks associated with each dataset are considered to be different, this approach can be categorized as multitask learning [47].

2.2. Transfer learning

Transfer learning aims at improving the performance of target learners or solving new problems faster on target domains by transferring the knowledge learned from different (but related) source domains [47]. This field of machine learning has applications in various real-world problems limited by the amount of training data [48], such as fault diagnosis [49] and medical imagery [50]. A condition for the transfer of knowledge is the existence of a similarity between tasks. For instance, the solar forecasting algorithm monitoring a new solar site, where local data are limited, could benefit from training on datasets generated in other locations to improve its generalization skills. This specific transfer learning approach, termed domain adaptation, aims at adapting a learning algorithm to a new data distribution while keeping the task unchanged. This method could be especially beneficial for new solar facilities which have a limited amount of data. Other similar but distinct activities in solar forecasting are, for instance, PV power output *versus* solar irradiance forecasting, or cloud cover modeling from sky images *versus* satellite observations.

There are different ways to categorize transfer learning approaches. An approach is known as *homogeneous transfer learning* if the type of input variables (e.g., sky images) and labels (e.g., irradiance) are the same for source and target domains (location A and B). In contrast, if input variables (e.g., sky images *versus* satellite observations) or labels (e.g., irradiance *versus* PV output) are distinct, the approach is referred to as *heterogeneous transfer learning* [47]. Another review splits methods

into four groups: instance-based (instance weighting strategy), feature-based (creation of a new feature representation), parameter-based (the transferred knowledge is encoded at the parameter/model level) and relational-based (transfer of the relationship among the source data to the target domain) approaches [51]. Instead of learning a new task from scratch, a common practice in deep learning is to train for a new task such as solar forecasting [52] starting from standard neural networks such as VGGNet [53], ResNet [54], DenseNet [55], etc., pre-trained on large datasets, e.g. ImageNet [56]. Alternatively, self-supervised learning, a form of unsupervised learning that generates pseudo-labels from the data itself, offers a framework to pre-train a model on an unlabeled dataset close to the target dataset instead of on a generic dataset.

Another review paper [57] describes two domain adaptation approaches based on the consistency of the neural network architecture. Symmetrical training corresponds to methods involving identical model architectures for both the source and target datasets. This modeling approach is often used in feature-based transfer learning. In contrast, asymmetrical training enables layer sharing between different architectures which can be applied to instance-based domain adaptation.

Some studies have applied transfer learning techniques to image-based solar forecasting and related fields. For instance, in sky image segmentation, a data representation can be learnt on a large unlabeled dataset of sky images with tasks such as image reconstruction, clustering and classification, prior to fine-tuning the model on the target segmentation task with a smaller labeled dataset [58]. Models pre-trained on the sky image dataset outperform those pre-trained on ImageNet [56] or randomly initialized. Pothineni et al. [45] claimed the deep learning model they developed (KloudNet) can be used to improve the performance on other PV plants via transfer learning, but no quantitative results are presented in their article. In solar irradiance estimation from satellite data, a recent work applied transfer learning to a four-layer neural network [59]. The source domain consists of simulated data used to pre-train the model while real-world satellite and in-situ observations are used for fine-tuning the last four layers. Although some negative transfer is observed, the proposed transfer learning approach benefits some tasks such as daily downward short-wave radiation estimation compared to baselines trained on simulated or in-situ data only.

3. Dataset

3.1. Dataset overview

In this study, three datasets collected globally from three different locations with drastically different weather conditions are used to explore transfer learning approaches. These datasets are (1) the Stanford dataset [40], collected on the campus of Stanford University in California, US, which is characterized by long summers with mostly clear sky days and short winters with partly cloudy skies; (2) the SIRTa dataset [44], collected by the SIRTa Atmospheric Observatory in Palaiseau, France, which contains a majority of partly cloudy and cloudy sky conditions, and (3) the DEWA dataset, collected from the outdoor testing facility of the Dubai Energy and Water Authority (DEWA) in the United Arab Emirates (UAE), which has a clear sky in most seasons but with sandstorms occurring usually during dry summers. Among the three datasets, the Stanford and SIRTa datasets are open source and publicly accessible,² while the DEWA dataset

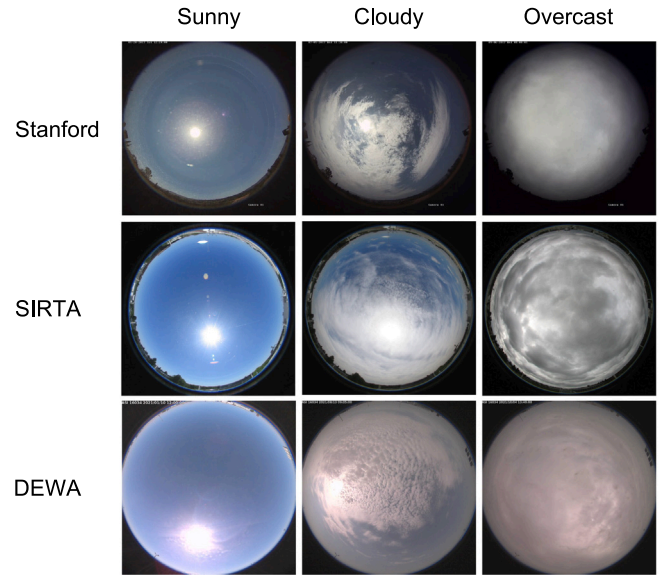


Fig. 3.1. Sky image examples from three locations with different weather conditions.

is private. It should be noted that only team Stanford has access to the DEWA dataset due to an internal collaboration contract between Stanford and DEWA, and team Cambridge only has access to the two publicly available datasets. The detailed comparison of these datasets is shown in Table 3.1.

The different weather conditions are illustrated by image samples collected at these three locations. Fig. 3.1 depicts sky images corresponding to different types of sky conditions: sunny, cloudy and overcast, respectively. It can be observed that the three cameras render sky images slightly differently in terms of the hue. For example, DEWA image samples appear yellowish and dusty due to the impact of sandstorms. In addition, different camera characteristics, e.g., camera contrast and saturation, can affect the image look.

3.2. Data processing

All the experiments conducted by both teams are based on the exact same processed datasets. The high-resolution raw images collected by the cameras are first down-sized to 64×64 pixels to save computation in the model training process. The PV output and irradiance measurements collected from the data loggers are averaged over a minute. To form valid samples for the forecasting task, each time point (t_0) is checked to ensure the availability of (1) the prediction target, i.e. the PV output/GHI measurements 15 min ahead ($t_0 + 15$); and (2) the model inputs, i.e. the sky images and concurrent PV output/irradiance measurements over the past 15 min at a 2 min resolution ($t_0 - 14, t_0 - 12, \dots, t_0$). All time stamps that do not satisfy these two conditions are dropped. For the Stanford dataset and the DEWA dataset, the sampling frequency is chosen to be 2 min (two samples differ in t_0 by 2 min) because higher frequency led to a longer model training time with limited improvement on the model accuracy [19]. For SIRTa dataset, the sampling frequency is set to be 1 min for 2017 and 2 min for 2018 to 2019 to be consistent with the imagery data frequency (see Table 3.1). After processing, the number of valid samples for the three datasets can be found in Table 3.1. In this study, the three datasets share similar characteristics such as similar camera orientations and their location in the same hemisphere. However, these aspects might partially hinder the transfer of knowledge in other contexts (e.g., different trajectories of the sun due to the location of the camera or its orientation). This could be addressed by the application of data augmentation techniques in the data processing stage. For instance, a transfer function independent of

² Detailed information about the Stanford dataset can be found via the following Github Repository <https://github.com/yuhao-nie/Stanford-solar-forecasting-dataset>; the SIRTa dataset is available upon request to https://sirta.ipsl.polytechnique.fr/data_form.html.

Table 3.1
Comparison of three studied datasets.

	Stanford	SIRTA	DEWA
Location	Stanford, US	Palaiseau, France	Dubai, UAE
Data type	Sky images & PV power output	Sky images & GHI	Sky images & GHI
Data frequency	1-min	1 to 2-min ^a	1-min
Image resolution ^b	2048 ² → 64 ²	650 ² → 64 ²	1920 ² → 64 ²
Time window	2017.3–2019.10	2017.1–2019.12	2021.1–2021.11
Camera model	Hikvision DS-2CD6362F-IV	EKO SRF-02	EKO ASI-16
Camera orientation	14° south by west	2° south by east	Due south
PV system	30-kW rooftop system with elevation 22.5°, azimuth 195°	N/A	N/A
Number of days	522	953	269
Number of valid samples after processing	135,527	448,268	91,979
Training, validation and test set split	84%:9%:7%	88%:10%:2%	83%:9%:7%

^a For the image data, the frequency is 1-min in 2017 and 2-min in 2018 to 2019; for the irradiance data, the frequency is 1-min for 2017 to 2019.

^b → means image down-sizing from the high resolution raw images.

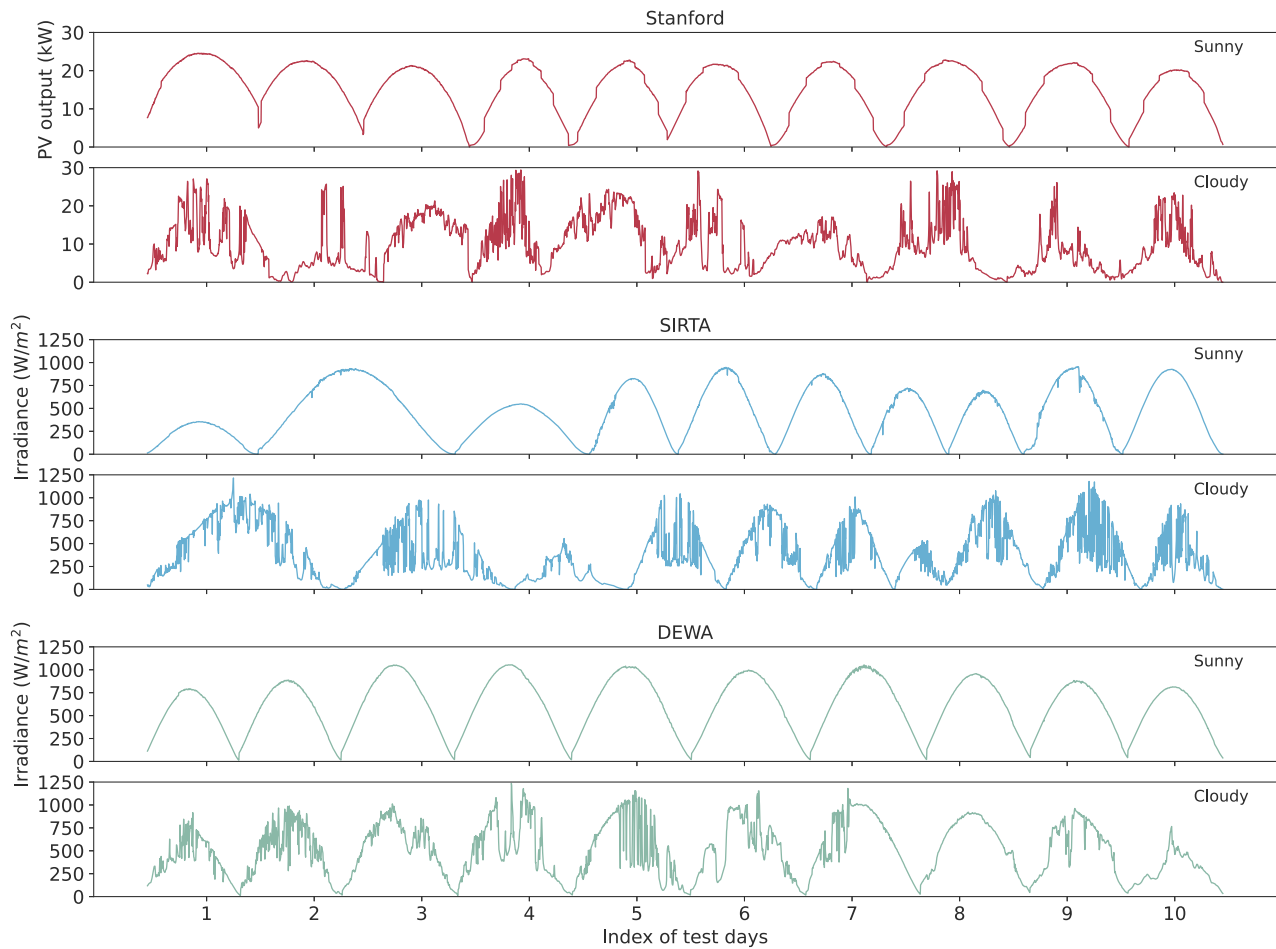


Fig. 3.2. The test set PV/irradiance profiles for the three datasets (The discontinuities visible in the sunny days of the Stanford test set are due to missing data points).

the camera orientation can be learnt by randomly rotating sky images during training [35] or by representing the scene with polar coordinates centered on the sun [60] using an image-based sun tracker [61].

3.3. Dataset partition

For model development and evaluation purposes, the valid samples of all three datasets are partitioned into the model development set (consisting of training and validation) and the test set. The test set is first separated out with 10 sunny days and 10 cloudy days across the entire time period and is never touched during the model development

processes. The PV output/irradiance profiles of these 20 days for all three datasets are shown in Fig. 3.2. The remaining data go to the model development set. Fig. 3.3 presents the model development set data distribution of the three locations. It can be observed that Stanford PV and DEWA irradiance distributions share some similarity while SIRTA irradiance distribution shows a nearly opposite trend compared to the other two datasets. To avoid the bias from data partitioning and obtain a less optimistic estimate of the model performance, ten-fold cross-validation is employed in this study, which divides the development set into 10 folds, 9 folds for training the model and 1 fold for validating the model. The model is trained 10 times, each time with a

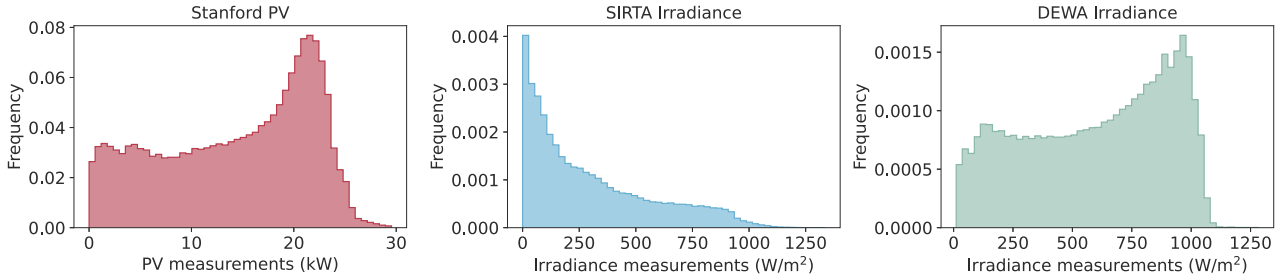


Fig. 3.3. The model development set data distribution of the three locations.

different fold as the validation set, resulting in 10 sub-models. The final prediction made by the model is the ensemble mean of these 10 sub-models. Under this setup, the split of training, validation and testing samples in percent for each dataset is shown in Table 3.1. Moreover, to avoid the data from the same day ending up in both training and validation sets, which could potentially lead to over-estimates of the model performance due to the closeness between training and validation samples, day-block shuffling is performed during the cross-validation process [19].

4. Methodology

The specific solar forecasting task tackled here is to predict 15-min-ahead PV power output or GHI values based on the three datasets described in Section 3. Two different deep learning models utilized by the two teams are presented in this section, including the model architectures and training details. Additionally, the metrics used to evaluate the performance of the models are also described. It should be noted that, this study does not aim at developing a new deep learning model, rather it focuses on comparing the effectiveness of different training strategies using existing solar forecasting models, namely, SUNSET [19] and ConvLSTM [28], which have already been compared with existing methods in original publications. The readers are referred to these two published articles for more details in terms of methodology comparison with existing methods. Meanwhile, to avoid the bias caused by particular model architectures and ensure the reliability of the results for understanding the impact of diverse training data, most of the experiments in this study have been conducted using both model architectures mentioned above.

4.1. Deep learning models

While the two teams utilized different deep learning models for the forecasting task, the following common setups are shared: (1) using hybrid features as the model input — sky images and measurements history (PV output/GHI) in the past 15 min with 2-min interval; (2) minimizing the mean squared error (MSE) loss function and using the stochastic gradient descent optimizer Adam [62] for model training; (3) employing 10-fold cross-validation and training 10 sub-models; and (4) for model evaluation, the prediction is generated by the ensemble mean of the predictions from the 10 sub-models. An illustration of the model architectures of both teams can be found in Fig. 4.1 and the comparisons of the architectures as well as training settings are listed in Table 4.1 with details described individually in the subsections below.

4.1.1. Stanford SUNSET model

A CNN architecture named SUNSET (Stanford University Neural Network for Solar Electricity Trend) is used by team Stanford. The SUNSET model is first introduced by Sun et al. [19] to forecast 15-min-ahead PV output and is characterized by its usage of CNN and hybrid input features. The 1-min lag term interval used in the original SUNSET model is changed to 2-min in this study and modifications were made

Table 4.1

Comparison of the two deep learning architectures and training details.

	SUNSET	ConvLSTM
Input sky images	Past 8 frames	Past 8 frames
Input PV or GHI history	Past 8 values	Past 8 values
Activation functions	ReLU	ReLU
Normalization layers	Batch normalization	None
Pooling layers	Max pooling	None
Dropout rate	0.4	0.0
Number of parameters	13.66M	4.25M
- Number of convolutional layers	2	7
- Number of residual blocks	0	6
Optimizer	Adam	Adam
Training loss	MSE	MSE
Learning rate	3×10^{-6} ^a / 2.5×10^{-5} ^b	1×10^{-5}
Batch size	256	10

^a For training on Stanford dataset.

^b For training on SIRTA dataset.

accordingly to accommodate the different input and output for each dataset, either PV or irradiance measurements.

The basic structure of the SUNSET model includes two Convolutional (Conv.) blocks and one Fully-connected (FC) block. The Conv. block employs a sandwich-like structure, including sequentially one Conv. layer, one batch normalization (BatchNorm) layer, and one pooling layer. The Conv. layer utilizes a 3×3 filter, with a stride of 1 and same-value padding. The activation function used for the Conv. layer is a rectified linear unit (ReLU). In the pooling layer, 2×2 max pooling with a stride of 2 is used to reduce the activation spatial dimensions. The first Conv. block contains 24 filters, while the second contains 48 filters. After the two Conv. blocks, the processed input is vectorized and concatenated with PV output/irradiance history and passed through the FC block to produce the prediction. The FC block includes two dense layers, each containing 1024 neurons and using ReLU as its activation function. After each dense layer, a dropout layer with a 0.4 dropout rate is performed to prevent over-fitting.

Different learning rates are used for model training on different datasets by team Stanford. For the Stanford dataset, a learning rate of 3×10^{-6} is used as this was seen to be effective via prior studies [19], while for the SIRTA dataset, a learning rate of 2.5×10^{-5} is used based on initial hyper-parameter selection experiments. Batch size is consistently set to be 256 for the stochastic gradient descent optimizer. An early stopping scheme is applied to prevent potential over-fitting, namely, the training is stopped when the validation loss is not observed to decrease for five consecutive epochs.

4.1.2. Cambridge ConvLSTM model

A ConvLSTM model architecture is used by team Cambridge, which was first presented by Paletta et al. [28] to forecast 10-min ahead GHI levels from past sky images and auxiliary data including irradiance measurements and the sun position. Several modifications were made to the original model architecture for this study. The first convolutional layer with a stride of two was changed to a layer with unitary stride to account for the lower spatial resolution of images in the present

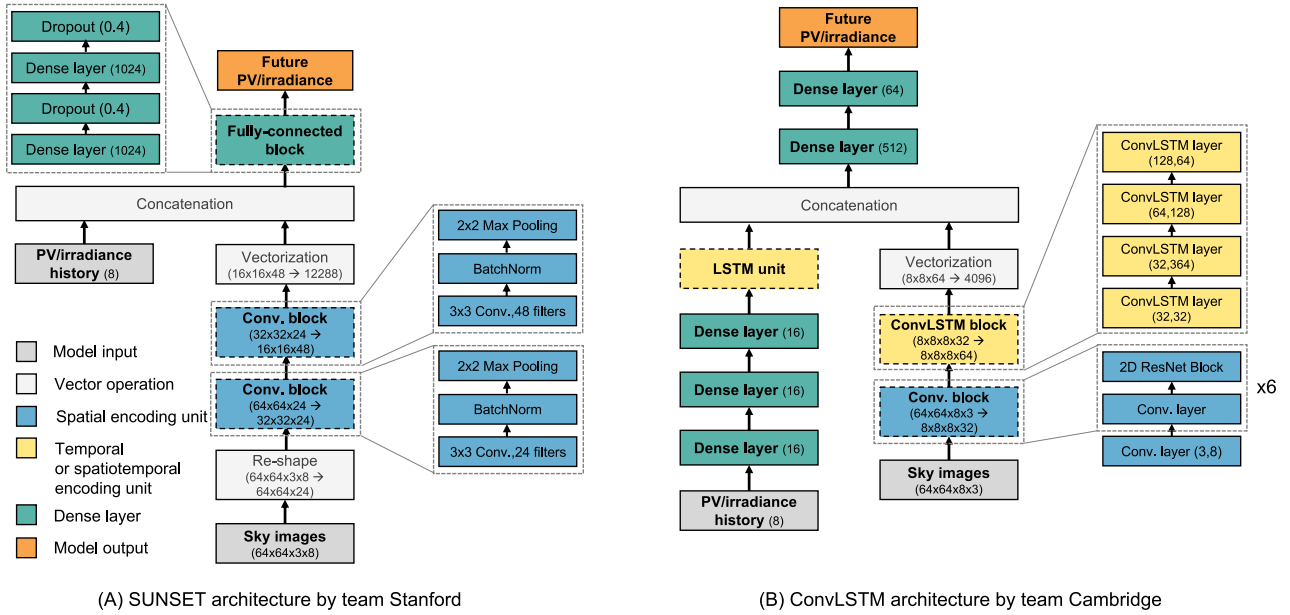


Fig. 4.1. Model architectures used by team Stanford and Cambridge.

work (64 pixels instead of 128). In addition, the model now takes three-channel images (RGB) as input instead of two (grey-scale short and long exposure images). Furthermore, the new model is fed only with the past GHI or PV output measurements as auxiliary input.

The ConvLSTM model is made of two parallel encoders for auxiliary data and sky images. Features from past irradiance or PV measurements are extracted with three dense layers and an LSTM layer [63]. Sky images are filtered through a sequence of convolutional layers and residual blocks [64] to decrease the spatial resolution from 64×64 to 8×8 . The resulting set of feature maps is then sequentially fed into a ConvLSTM module [65]. A learning rate of 1×10^{-5} is used to train the ConvLSTM model for both datasets with a batch size of 10.

4.2. Evaluation metrics

In this study, the model performance is evaluated from two aspects: (1) the prediction accuracy, which is measured by some common error metrics via applying the trained models to the test sets; and (2) the training cost, which reflects the consumption of computational resource during the model training.

To assess the prediction accuracy, the error metric root mean squared error (RMSE) is used. RMSE is the most commonly used metric and can be expressed by Eq. (1); other similar metrics like mean absolute error (MAE), mean bias error (MBE), etc. are not covered in this study.

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (\hat{Y}_i - Y_i)^2} \quad (1)$$

where N is the number of samples, $\hat{Y}$ is the prediction generated by the model and Y is the ground-truth measurement.

The forecasting skill (FS) is another common approach in solar forecasting evaluation, which essentially quantifies how much better/worse the error (usually in RMSE) of the proposed model is compared to that of a reference model (Eq. (2)).

$$\text{FS} = \frac{\text{RMSE}_{\text{ref}} - \text{RMSE}_{\text{forecast}}}{\text{RMSE}_{\text{ref}}} \quad (2)$$

The reference model mostly used for computing FS is the smart persistence model (SPM), which assumes the relative output stays constant from time t_0 to $(t_0 + T)$ (Eqs. (3) and (4)).

$$\hat{Y}(t_0 + T) = k_{\text{clr}} Y_{\text{clr}}^*(t_0 + T) \quad (3)$$

$$k_{\text{clr}} = \frac{Y(t_0 + T)}{Y_{\text{clr}}^*(t_0 + T)} = \frac{Y(t_0)}{Y_{\text{clr}}^*(t_0)} \quad (4)$$

where k_{clr} represents the relative output, or formally named as clear sky index, and Y_{clr}^* is the theoretical irradiance value or PV output estimated by a clear sky model. For estimating the clear sky irradiance for each time stamp in SIRTa dataset, the Helioclim database is used [66]. For Stanford dataset, the clear sky PV output is estimated using a canonical geometric irradiance model [67] as used by Sun et al. [19], shown by Eq. (5):

$$P_{\theta}(t) = P_s A_e \{ \cos \epsilon \cos \chi(t) + \sin \epsilon \sin \chi(t) \cos[\xi(t) - \zeta] \} \quad (5)$$

Where P_s is the maximum solar irradiance, 1000 W/m^2 ; A_e is the effective PV panel area, 24.98 m^2 , which is obtained from a least square fit with the real panel output of 12 clear sky days (details can be found in study by Sun et al. [19]); ϵ and ζ are elevation and azimuth angles of the solar PV arrays, which are 22.5° and 195° , respectively; $\chi(t)$ and $\xi(t)$ are the zenith and azimuth angle of the sun, which can be estimated for any minute of the year from the empirical functions provided in the textbook by da Rosa [67].

Given the scope of this study, other metrics such as the ramp score, temporal distortion metrics and parity plots [68] are not evaluated here. These metrics have been presented in original publications of the proposed models, namely, SUNSET [19] and ConvLSTM [28]. The readers are encouraged to check the original publications for details about other evaluation metrics.

For evaluating the training cost, a metric called training effort (TE) is defined, which is essentially the total number of samples seen by the model until convergence and can be expressed by Eq. (6):

$$\text{TE} = \text{Number of training epochs} \times \text{Training set size} \quad (6)$$

The number of epochs is obtained when the model stops training. Specifically, team Stanford uses an early stopping scheme for the SUNSET model, and the training is stopped when the validation loss does not decrease for five consecutive epochs. Team Cambridge stops the training when the ConvLSTM model starts overfitting, i.e. when the validation loss increases for several consecutive epochs. The training time is not used for comparison in this study, as it varies with the different GPU models used in training (Stanford used Tesla A100, while Cambridge used GeForce GTX 1080).

5. Experiments

This section details the experiments conducted to address the questions raised in the Introduction section. First, the local models are trained individually based on the dataset from each location, which serve as the baselines for comparison with other models developed in this study. The approaches used for baseline model development described in [19,28] correspond to the standard methodology in the literature. Next, a global model is trained jointly based on the combination of datasets from different locations. In view of the sample heterogeneity of different datasets, the effect of different input and output normalization methods and model architectures to accommodate input data streams from different locations are investigated. Moreover, the potential of knowledge transfer between different datasets via transfer learning is explored.

5.1. Training local models

The two research teams first train their models individually on each one of the three datasets to construct a local baseline. To examine the generalization of the local models, in other words, to test how well the local models perform when applied offsite without re-training, the models are trained based on the normalized data instead of the original data. Training on normalized data enables the models to be deployed across different locations with different scales or types (e.g., trained on irradiance values but applied to predict PV data) of the prediction targets. Specifically, during the development phase, the models learn to predict the relative values of PV power output or irradiance, and during the implementation phase, the relative predictions generated by the models are post-processed to revert them to the original scale. Here, the data normalization method used by each team is the optimal normalization method identified when training models on integrated datasets. Details can be found in Sections 5.2 and 6.2.

5.2. Training global models

The challenges of training global models are associated with the heterogeneity of samples from different datasets. A common problem is the different scales (e.g., 30-kW PV system versus 2-MW PV system) and/or different types (e.g., irradiance versus PV output) of the measurement data. This work mainly focus on the second problem. Moreover, differences in the camera setups or data distribution caused by the unique local weather conditions, could add other layers of difficulty. The PV output data in the Stanford dataset and the irradiance measurements in the SIRTA dataset are a good example demonstrating all of the above issues (see Fig. 3.3), and thus will be our main focus in this section. Other possible combinations of datasets (e.g., Stanford+DEWA, Stanford+SIRTA+DEWA) could essentially go through similar training processes. Specifically, to combine the two datasets for joint training, using different normalization methods for processing the model input and output data was explored. Tuning the model architectures to better accommodate the multiple input data streams was also experimented.

5.2.1. Input and output normalization

To be clear, the normalization in this study is defined in the form of $(X - a)/b$, where X is the data to be normalized, and a , b are normalization factors. Two types of data are involved, namely, images and sensor measurements. For sky images, both teams normalized the pixel values of the images from $[0, 255]$ to $[0, 1]$ by dividing by the maximum pixel value 255, which is a common technique used for image data normalization. For PV output/irradiance measurements, including both input and output, different normalization methods are examined in this section and the details are summarized in Table 5.1. With ground truth prediction data being inaccessible when the model is deployed, the normalization factors are solely based on the input data, which is statistically similar. After the input and output of each dataset

Table 5.1

Normalization methods in the form of $(X - a)/b$ for Stanford and SIRTA datasets.

Normalization method	Stanford	SIRTA
$X/\max$	$a = 0, b = 30$	$a = 0, b = 1366$
$X/95\text{tile}$	$a = 0, b = 24$	$a = 0, b = 853$
$X/\text{std.}$	$a = 0, b = 8$	$a = 0, b = 270$
$(X - \text{mean})/\text{std.}$	$a = 15, b = 8$	$a = 301, b = 270$
$X/(\max/100)$	$a = 0, b = 0.3$	$a = 0, b = 13.36$
$X/(95\text{tile}/100)$	$a = 0, b = 0.24$	$a = 0, b = 8.53$
$X/(\text{std.}/100)$	$a = 0, b = 0.08$	$a = 0, b = 2.7$
$(X - \text{mean})/(\text{std.}/100)$	$a = 15, b = 0.08$	$a = 301, b = 2.7$

Notes: X represents the measurement values and b is referred to as the normalization factor; *std.* and *95%tile* represent standard deviation and 95 percentile of the data, respectively.

are normalized, they are combined for global model training. Fig. 5.1 shows the data distribution of both datasets after normalization. In the testing phase, the same normalization factors are applied to the input data of the test set and the predictions are post-processed to the original scales. The global models on the test sets of the two datasets are evaluated individually and compare them with the corresponding local models.

5.2.2. Architecture tuning

The multi-location aspect of the data remains a key challenge for data fusion and transfer learning applications. While the baseline model might be able to learn location-specific features implicitly from the data to adjust its predictions based on the location of interest, the possibility of encoding the location information more explicitly motivates the development of alternative modeling strategies via slight modifications of the model architecture. Two additional methods are therefore proposed and compared with the baseline architectures described in Section 4.1. Fig. 5.2 illustrates the two alternative architectures, which are named as Architecture 1 [Arch. 1, see Fig. 5.2(B)] and Architecture 2 [Arch. 2, see Fig. 5.2(C)]. The modifications from the baseline architecture are highlighted with yellow background. It should be noted that here the SUNSET model is used to explain the architecture modifications, while the same idea can be applied to the ConvLSTM model. The models are notably trained with the normalized data based on the optimal normalized method identified for each team via experiments described in Section 5.2.1.

Arch. 1 has minor modifications compared with the baseline. The only change is the introduction of a condition matrix, with the same resolution as the sky images (64×64). All elements of the condition matrix are either 1 or 0, indicating the location of the data, for example, if the sky images come from the Stanford dataset, the condition matrix elements are all 0s, whereas if the sky images come from the SIRTA dataset, the condition matrix elements are all 1s.

Arch. 2 adds more complexity to the baseline architecture. It has shared convolutional blocks and two separate fully-connected blocks, with the two fully-connected blocks consisting of the same components as the baseline architecture [see Fig. 4.1(A)]. The shared convolutional blocks act as a common feature extractor and learn features from both locations, while the two separate fully-connected blocks learn the correlation between the image features and the PV output/irradiance individually for each location, which is based on the fact that the sun angle trigonometry changes with locations. During the training process, each of the fully-connected blocks will generate a prediction and the two predictions are then stacked into a vector. To generate the final prediction, an inner product is computed between the prediction vector and the one-hot label vector, which indicates the location of the data, with $[1, 0]$ and $[0, 1]$ representing the Stanford and SIRTA data, respectively.

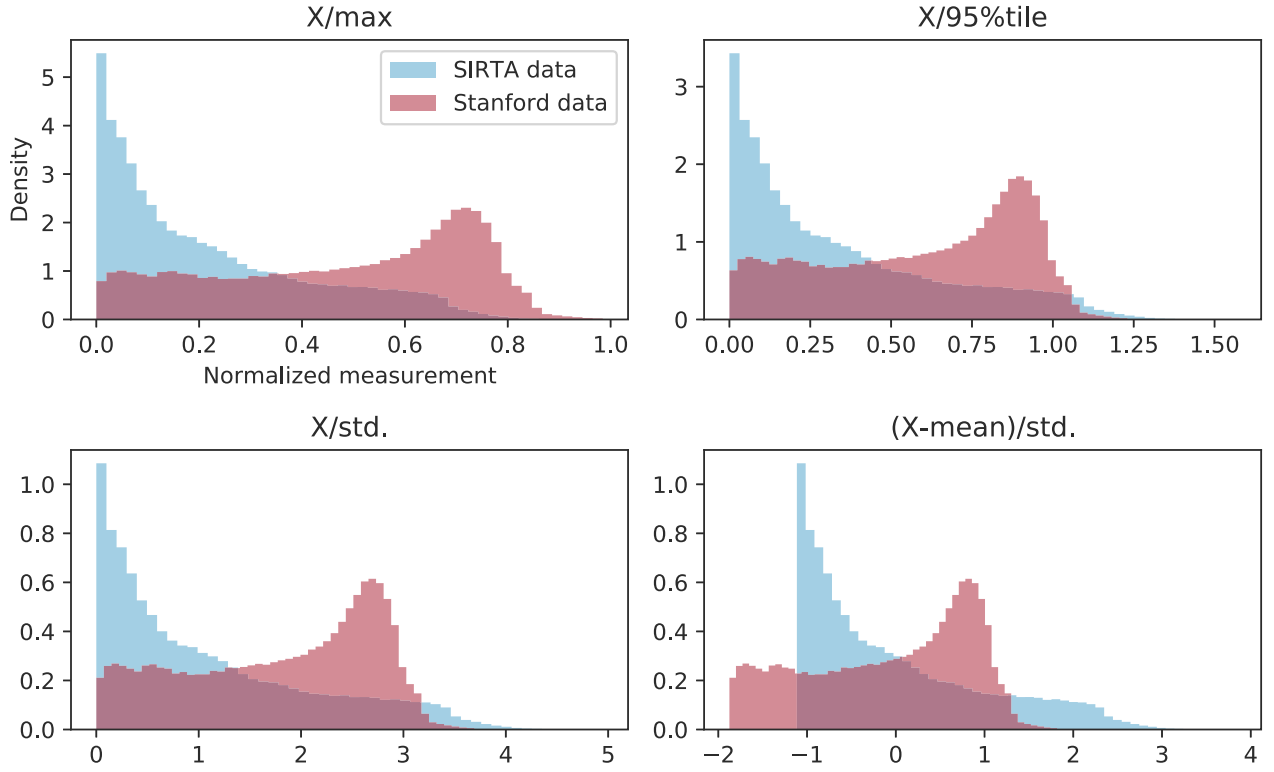


Fig. 5.1. The distribution of normalized measurements of the Stanford SIRTa datasets (Note: the distributions of $X/(max/100)$, $X/(95\%tile/100)$, $X/(std./100)$, $(X-mean)/(std./100)$ are not shown here because they have similar distributions as their counterparts and the only difference is the scale of 100x).

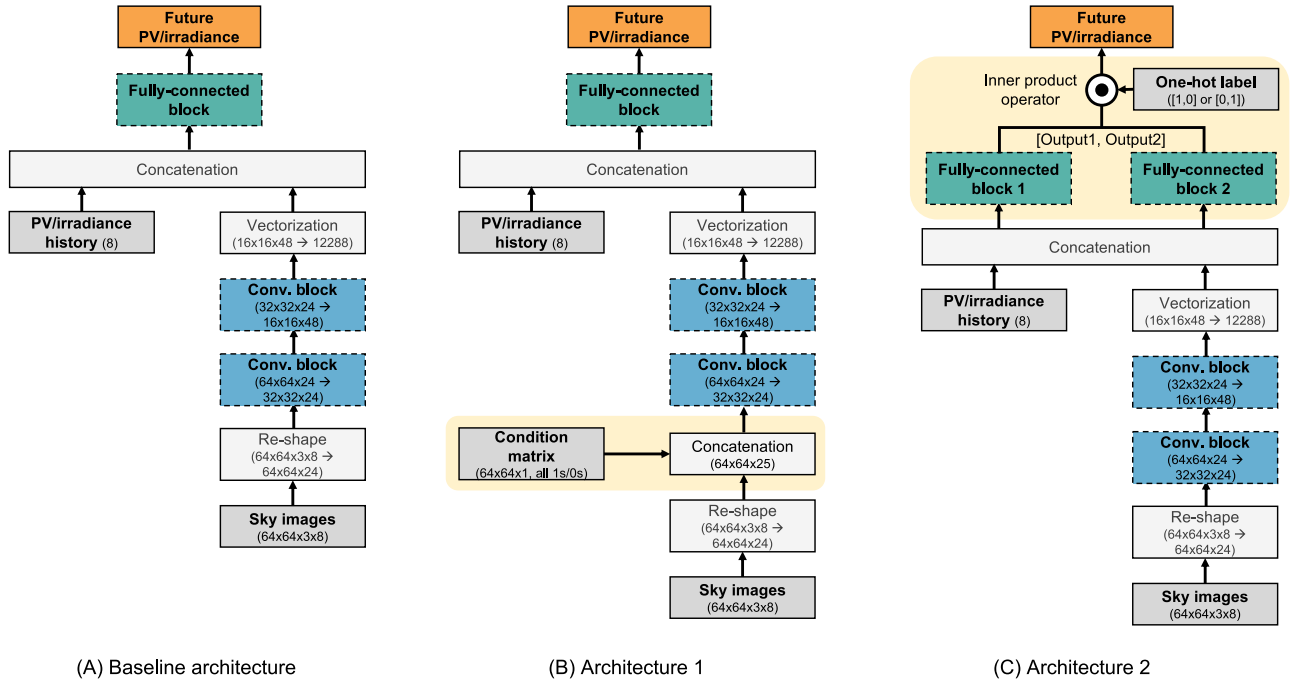


Fig. 5.2. Illustration of different architecture set-ups using the SUNSET model as an example (the modifications from the baseline architecture are highlighted with yellow background).

5.3. Transfer learning

This section further examines if the knowledge, or specifically the feature representations from a pre-trained solar forecasting model can be leveraged, so that the efforts to train a model from scratch in a new location of interest could be avoided. Here, a source dataset D_S

and a corresponding source model $\mathcal{M}_S$, as well as a target dataset D_T and a target model $\mathcal{M}_T$ are defined. In most cases, D_S is a large dataset that contains a large quantity of samples, whereas D_T is limited in the number of samples. The goal of transfer learning, specifically, domain adaptation, is thus to learn $\mathcal{M}_T$ using D_T with the knowledge gained from learning $\mathcal{M}_S$ based on D_S . It should be noted that the

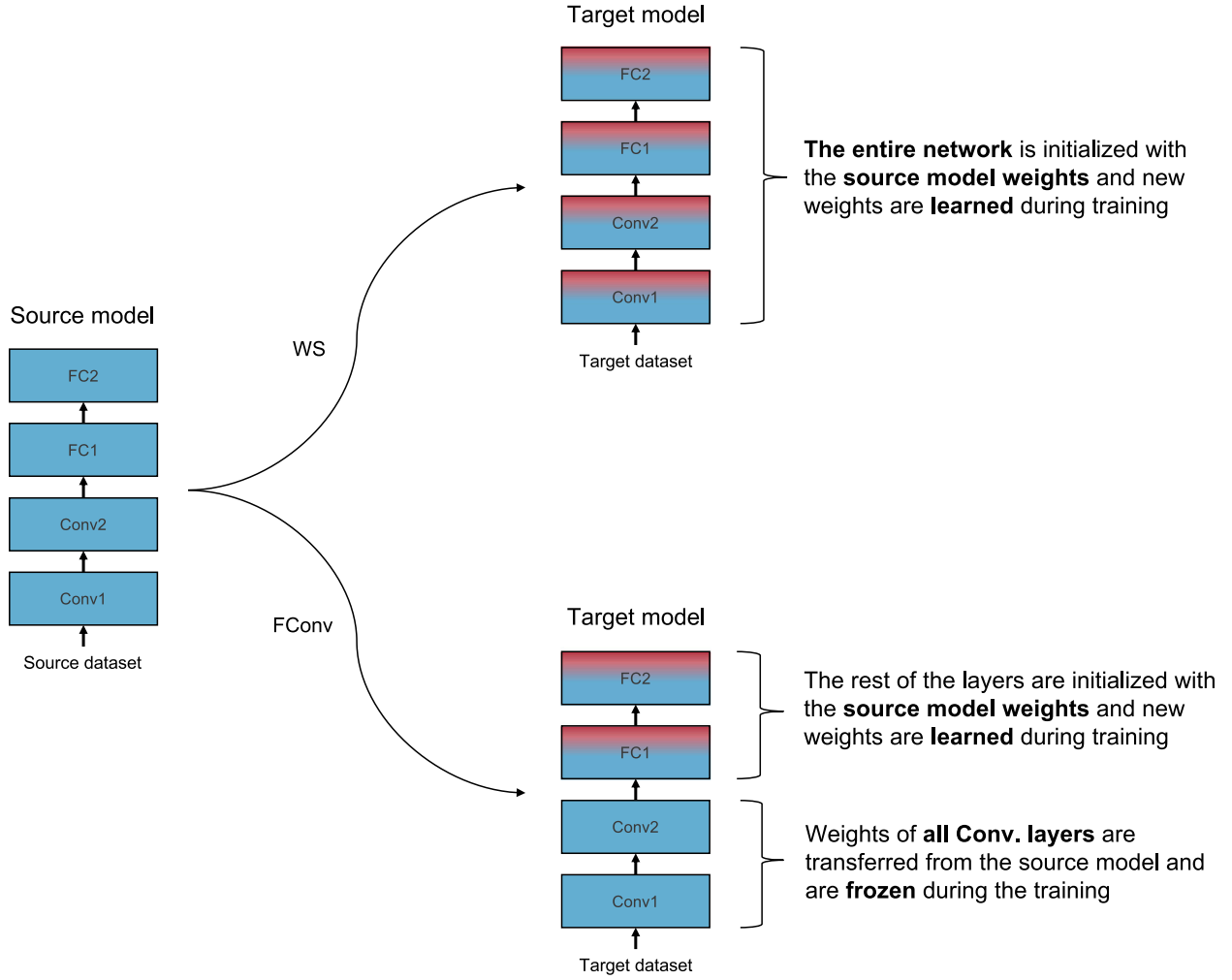


Fig. 5.3. Illustration of different transfer learning strategies examined in this studies, taking a CNN with two convolutional layers (Conv1 and Conv2) and two fully connected layers (FC1 and FC2) as examples.

same architecture is used for $\mathcal{M}_S$ and $\mathcal{M}_T$ in this study, namely, either SUNSET by team Stanford or ConvLSTM by team Cambridge.

To implement transfer learning, $\mathcal{M}_S$ is first developed based on D_S . Two different strategies are then investigated for transferring the knowledge from $\mathcal{M}_S$ to $\mathcal{M}_T$:

- warm-starting strategy (WS): instead of initializing the weights of $\mathcal{M}_T$ randomly, it initializes the entire network with the weights of $\mathcal{M}_S$, and from that initial point, the training on D_T is started and new weights are learned;
- freezing Conv. blocks strategy (FConv): the weights for all Conv. blocks from $\mathcal{M}_S$ are transferred and are frozen during the training process, while the weights of fully-connected blocks are initialized with the weights from $\mathcal{M}_S$ and learned with the target dataset D_T .

For FConv strategy, the feature extractors of $\mathcal{M}_S$ are reused by $\mathcal{M}_T$ and only the mapping functions associated with the fully connected layers are learned based on new data. An illustration of these two transfer learning strategies can be found in Fig. 5.3.

Given that D_S generally has a wider data coverage than D_T , the DEWA dataset is first used as D_T , and experiment with the following combination of transfer learning: $\{D_S: \text{Stanford} \rightarrow D_T: \text{DEWA}\}$, $\{D_S: \text{SIRTA} \rightarrow D_T: \text{DEWA}\}$ and $\{D_S: \text{Stanford+SIRTA} \rightarrow D_T: \text{DEWA}\}$. The possibility of knowledge transfer between two sky image datasets with drastically different prediction targets and data distribution, namely, $\{D_S: \text{SIRTA} \rightarrow D_T: \text{Stanford}\}$ and $\{D_S: \text{Stanford} \rightarrow D_T: \text{SIRTA}\}$, is

then examined. Different amounts of training data for D_T , including 1%, 5%, 10%, 20%, 50%, 75% and 100%, roughly corresponding to 4, 16, 29, 55, 124, 184, 249 days of data for the DEWA development set (starting in January), 6, 29, 59, 118, 272, 390, 502 days for the Stanford development set (starting in March), and 9, 42, 79, 133, 329, 631, 933 days for the SIRTA development set (starting in January) are also experimented. The data are sampled chronologically in the whole dataset to mimic the real situation of data collection. It should be noted that as team Cambridge does not have access to the DEWA dataset, they only conduct the second part of experiments and team Stanford (with access to the DEWA dataset) conducts all the experiments mentioned above. In this study, both the source models $\mathcal{M}_S$ and the target models $\mathcal{M}_T$ are trained using the normalized data rather than the original data, again based on the optimal normalization method identified by each team. All transfer learning models are evaluated and compared with the local model baseline trained from scratch using D_T in two aspects: prediction accuracy measured by the RMSE, and training cost measured by the TE as described in Section 4.2.

6. Results and discussion

6.1. Comparison of local and global models for solar forecasting

The performance of local and global solar forecasting models are first compared. Besides serving as baselines, the local models are also applied offsite without re-training by predicting the relative values of

Table 6.1

Overall test set performance of local and global models (the best prediction performance on each local test set is highlighted in bold font).

Model	Trained on	Training epochs	Training samples	Training effort	Test RMSE (↓) on		
					Stanford	SIRTA	DEWA
SUNSET	Stanford	21.2 ± 6.1	125,876	2.67 ± 0.77M	2.68	136.96	155.82
	SIRTA	9.6 ± 1.8	438,172	4.21 ± 0.81M	6.53	96.42	219.79
	DEWA	11.8 ± 2.2	85,278	1.01 ± 0.19M	4.65	160.28	89.49
	Stanford+SIRTA	28.6 ± 5.6	564,048	16.13 ± 3.15M	2.61	96.34	129.21
ConvLSTM	Stanford	11.7 ± 2.7	125,876	1.47 ± 0.34M	2.62	113.05	N/A
	SIRTA	5.6 ± 2.6	438,172	2.49 ± 1.13M	4.39	96.46	N/A
	Stanford+SIRTA	6.0 ± 1.7	564,048	3.36 ± 0.99M	2.62	95.53	N/A

Notes: (1) The training effort reported in this table is defined as the number of samples seen by the model during training until convergence, i.e. the number of training epochs × the number of training samples, representing the average performance (mean ± std) for training one sub-model from ten-fold cross-validation. M stands for million samples. (2) The training epochs are obtained when the model stops training, namely, the validation loss is not observed to decrease for five consecutive epochs (3) The test RMSE is calculated based on the ensemble mean prediction of the ten sub-models from ten-fold cross-validation.

irradiance/PV output and post-processing to the original scales for performance evaluation. A global model is trained using the combination of Stanford and SIRTA dataset with normalized PV and irradiance values. The normalization methods used for both local models and global models are $X/(std./100)$ for team Stanford and $X/95\%tile$ for team Cambridge, which are the optimal normalization methods identified in the experiments presented in Section 6.2.

Table 6.1 shows the overall test set performance of local and global models trained with baseline SUNSET and ConvLSTM architectures. For both architectures, the local models achieve the best performance when applied locally, and the prediction errors significantly increase when applied offsite (without re-training). Although there are still large prediction errors when applying local models to other locations with similar measurement data distributions, e.g., applying the local model developed using Stanford data (PV output) to the DEWA test set (GHI) (Test RMSE 155.82 W/m²), it tends to have better performance than applying local models to locations with quite different data distribution even though they share the same target variable, e.g., applying the local model developed using SIRTA data (GHI) to DEWA test set (GHI) (Test RMSE 219.79 W/m²).

The prediction errors of applying learned local models offsite can be largely attributed to two parts of the model architectures, the feature extraction part associated with the Conv. blocks and the regression part associated the fully-connected blocks. It is found that the errors mainly come from the regression part, namely, mapping extracted features with the prediction target values, whereas the feature representations learned based on each individual dataset can to some extent be shared. This is illustrated in Fig. 6.1 which shows the predictions of local models developed based on Stanford and SIRTA datasets individually using the SUNSET architecture applied to 6 example days (3 sunny and 3 cloudy days) in the DEWA test set without re-training. The first column of the figure shows the original predictions, and the second column shows the scaled predictions obtained by simply multiplying the original predictions by a factor of 1.2 for the Stanford local model and 1.4 for the SIRTA local model for both example sunny and cloudy days, without any other treatment. The original prediction curves, including both Stanford and SIRTA local model predictions, tend to have similar trends and shapes as the ground truth curves, regardless of sunny days and cloudy days, which suggests that feature representations learnt by the models are common and can be shared between locations. The errors are mostly caused by the scale or magnitude of the predictions. Once the prediction is manually scaled by a factor,³ significant improvement in the prediction can be observed, although it is not perfect. This finding is further validated in Section 6.3 by freezing the weights of the Conv. blocks and only training the fully connected blocks during transfer learning.

³ Our further analysis finds that these factors might be related to the ratio of sun elevation angles for different locations.

In contrast, the global model trained with a combined dataset from two locations can perform well on individual test sets from both locations. The performance is close to, or even better than, the corresponding local models (e.g., Stanford+SIRTA global model *versus* Stanford local model *versus* SIRTA local model), which suggests that the global models can learn features from both locations simultaneously and the learned features can be correlated with the prediction targets in a relatively separate fashion without compromising the performance for each location. This property is further examined in Section 6.2 by comparing the baseline architectures with the modified architectures that explicitly ingest the location information. Moreover, including diversified samples from both locations could improve the model generalization ability.

Although applying Stanford+SIRTA global model to the DEWA test set gives much worse performance than the DEWA local model, it is significantly better than the Stanford and SIRTA local models applied to the DEWA test set. It could be reasonably expected that training a global model on a combination of Stanford, SIRTA and DEWA sets could give promising results on each of the three datasets.

Training a global model with a combined dataset and applying it to locations of interest achieves better prediction accuracy than training individual local models, especially if the local datasets have limited sample size. However, the training cost of global models remains a challenge. Depending on specific model architectures, more efforts might be required in the training process, due to increases in both training set size and training epochs, for the model to converge. A further evaluation on the training cost is presented in Section 6.4 for comparison of local, global models, and models trained with transfer learning.

6.2. Dataset fusion and architecture tuning for global solar forecasting models

In view of the superior prediction accuracy of the global models over the local models, this section explores the impact of the heterogeneous scales and distributions of the target variables on the training of global models. In addition, two alternative model architectures designed to better accommodate the multi-location input data are investigated and compared with the baseline model architectures.

Fig. 6.2 shows the comparison of different normalization methods for training the global model for SUNSET and ConvLSTM architectures, respectively. The global model is trained jointly using the combination of Stanford and SIRTA datasets and evaluated individually on the two local datasets. It can be observed that the SUNSET model is more sensitive to the scale of the normalization factors compared with the ConvLSTM model, dividing the normalization factor by 100 appears to significantly improve the model performance on both local datasets. In contrast, a small normalization factor generally leads to a worse performance of the ConvLSTM model. The optimal normalization factors for SUNSET and ConvLSTM are identified as $(0, std./100)$ and $(0, 95\%tile)$,

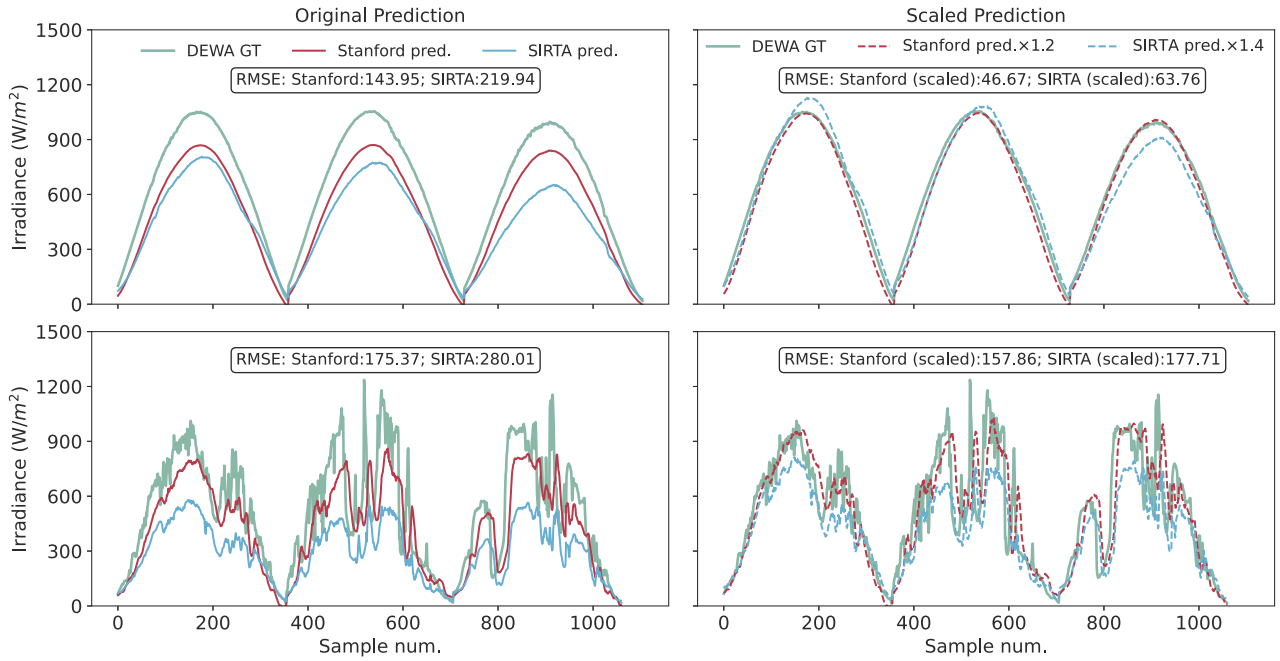


Fig. 6.1. Visualization of Stanford and SIRTa local model predictions based on the SUNSET architecture applied to 6 example days (3 sunny days shown in the first row and 3 cloudy days shown in the second row) in the DEWA test set without re-training. The first column shows the original predictions and the second column shows the scaled predictions from simply multiplying the original prediction by a factor. The same factor is applied for both sunny days and cloudy days predictions. (GT: ground truth; pred.: prediction).

respectively. It should be noted that normalization factors derived from robust statistics, e.g. *std* or *95%tile*, are less affected by outliers compared to, say, the *max* value. Preliminary experiments suggest that the different responses to the normalization factors might be due to particular architectural design choices of the two models, especially the batch normalization layers (SUNSET uses batch normalization layers but ConvLSTM does not use), though the underlying reasons need to be more thoroughly understood via future studies. It is also suggested that researchers should take care in deciding which normalization methods they use for their models.

The empirical results of ingesting location information in the model architecture (see Fig. 5.2 for a comparison of different global model architectures) are presented in Table 6.2. Compared with the baseline architectures, no significant performance improvement is found for the two alternative architectures on two local test sets for both SUNSET and ConvLSTM models. For example, by using Arch.1, FS only increases slightly from 18.6% to 18.8% for SUNSET on Stanford test set, while it deteriorates considerably from 18.3% to 13.9% for ConvLSTM. This suggests that the baseline model is able to distinguish location using subtle features within the sky images or past solar measurements. Although the modifications in the architecture to condition the predictions on the location or disentangle the prediction for each location does not provide a clear performance gain when applied to two datasets, a modeling scenario involving more databases or similar cameras might benefit from such a guidance and need more exploration for future research. Alternatively, the use of the clear-sky index k_{clr} as a normalized variable could facilitate the integration of heterogeneous solar time series into a single modeling framework at the cost of potential instabilities in low solar elevation conditions.

6.3. Knowledge transfer for solar forecasting

Lastly, the potential of knowledge transfer from a solar forecasting model pre-trained on a source dataset by fine-tuning it on a target dataset is explored.

The first experiment explores transfer learning scenarios from diverse source training sets – Stanford, SIRTa and Stanford+SIRTa –

Table 6.2

Performance of different global model architectures on local test sets (the best performance is highlighted in bold font). The global models are trained on Stanford+SIRTa dataset and tested on Stanford and SIRTa dataset individually.

Model	Normalization	Architecture	Test RMSE (↓)/FS (↑) on	
			Stanford	SIRTa
SPM	N/A	N/A	3.21/0%	114.2/0%
SUNSET	$X/(std./100)$	Baseline	2.612/18.6%	96.335/15.6%
		Arch. 1	2.606/18.8%	96.075/15.9%
		Arch. 2	2.610/18.7%	96.585/15.4%
ConvLSTM	$X/95\%tile$	Baseline	2.622/18.3%	95.529/16.3%
		Arch. 1	2.765/13.9%	96.850/15.2%
		Arch. 2	2.663/17.0%	96.264/15.7%

to the DEWA dataset. Fig. 6.3 highlights the performance of the two transfer learning strategies (WS and FConv) relative to the baseline local training and for different amounts of target data as described in Section 5.3. The models used for this experiment are based on the SUNSET architecture and the solar forecasting performance is evaluated on the DEWA test set.

In general, models trained using transfer learning strategies, either WS or FConv, are more accurate than the DEWA local baseline models regardless of the share of target data used. The smaller the local dataset is, the more benefits transfer learning brings. When target data is less than 20% of the whole DEWA dataset (55-day-equivalent amount of data), the transfer learning models significantly outperform the local baseline from approximately 10% and up to 60%. Regarding the extreme case of training or fine-tuning models on only 1% of total data (4-day-equivalent amount of data), the transfer learning models outperform the baseline by nearly 60%. This indicates that the knowledge gained from training models on large datasets can be transferred and reused for a new task and at a new location for a jump-start. Although the potential benefits of transfer learning gradually diminish from 20% target data and beyond, it still outperforms the baseline with improvement gains within 10%.

In terms of different transfer learning strategies, namely, WS and FConv, the difference in prediction performance is generally less than

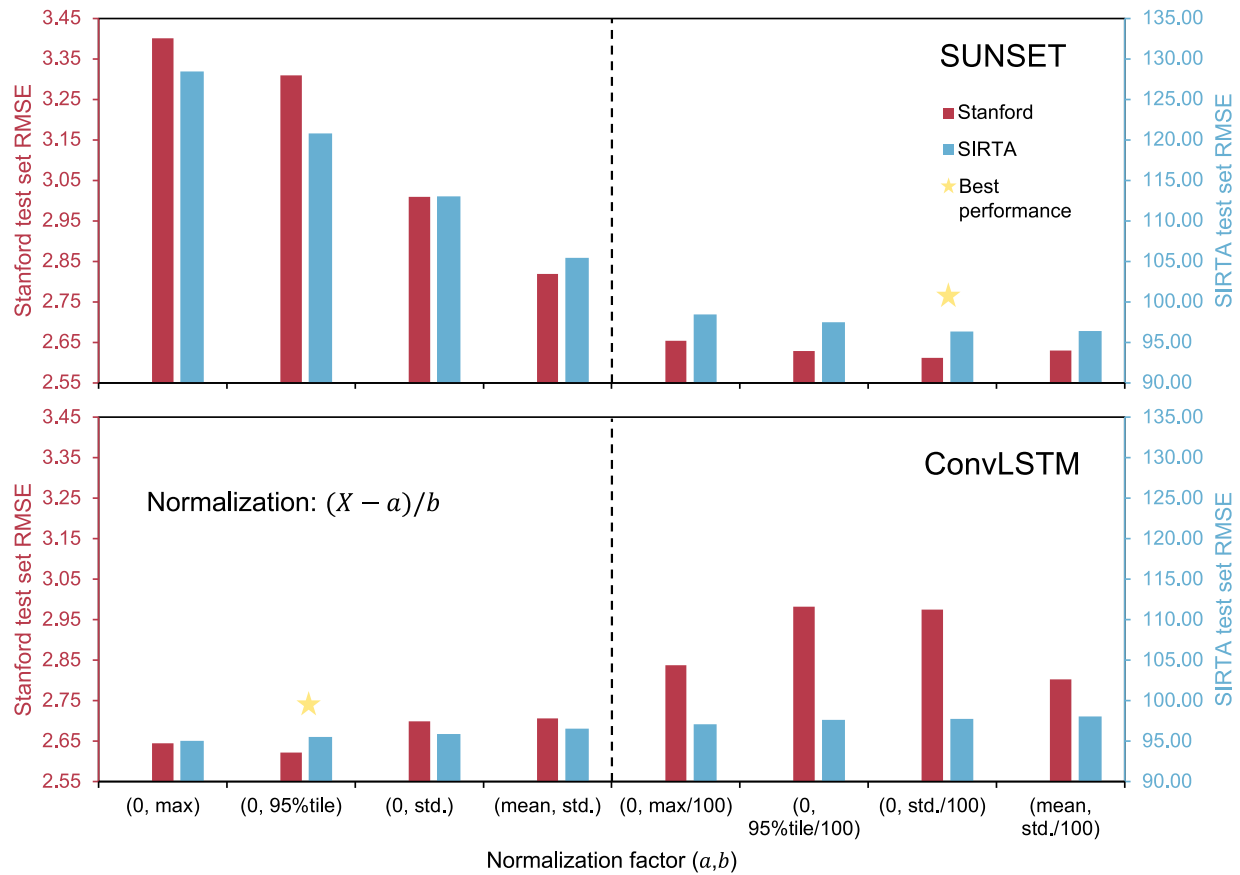


Fig. 6.2. Comparison of different normalization methods for training a global model for two different model architectures.

1%. Although in some specific situations such as low data availability (e.g., 1% target data), a slight difference in RMSE can be observed (less than 5%). Generally, there is no significant advantage for using one strategy over the other. To this end, freezing all the Conv. blocks during the model training is totally acceptable, as the feature representations learnt by the source models can largely be reused by the target model without compromising the performance. Even when the data distribution is drastically different, e.g., the SIRTA dataset *versus* the DEWA dataset, some common features learned by the source model can be shared.

Moreover, by comparing different source models for transferring the knowledge, it can be observed that the larger and the more diversified the dataset used for building the source model, the more beneficial the local fine-tuning via transfer learning. Using a combination of datasets such as the Stanford and SIRTA datasets to train a source model might not only provide a richer diversity of meteorological patterns to learn from, but might also prevent the model from overfitting to a particular location by learning more general common features. When the target data is less than 20% of the full dataset, the Stanford→DEWA model performs better than the SIRTA→DEWA model, most likely because the Stanford and DEWA datasets share a similar data distribution. As such, some of the learned features can be easily transferred. However, when the amount of target training data increases the opposite trend is observed. For instance, when target data is greater than 20%, the learning of the sun angle equation improves substantially (see Fig. 6.4) as shown by clear-sky day predictions almost aligned with the ground truth. Consequently, the overall performance of the model on the test set is predominantly determined by the solar forecasting accuracy during cloudy days. Therefore, the features learned from the SIRTA dataset, which is dominant in cloudy data, can contribute more. Moreover, transfer learning seems to provide a significant gain in terms of target data requirements. For example, it can be noticed that with

only 20% target data the Stanford+SIRTA→DEWA transfer learning model performs slightly better than the local baseline trained on 100% of the DEWA target dataset.

Even though the two datasets have very different data distributions, transferring knowledge from SIRTA to DEWA shows promising results as described in previous paragraphs. Here, two even harder cases, namely, transfer learning from Stanford to SIRTA and from SIRTA to Stanford, with differences in both prediction target (PV output *versus* GHI) and data distribution, are further examined. The Stanford dataset contains only approximately a quarter of the data volume of the SIRTA dataset, and has a relatively more balanced sunny and cloudy sample distribution than the SIRTA dataset, which is dominated by cloudy data. Fig. 6.5 depicts the solar forecasting performance of the SUNSET and ConvLSTM models trained via different transfer learning strategies between the Stanford and the SIRTA datasets and with increasing amounts of target data.

Overall, the error curves of the SUNSET and ConvLSTM models associated with both transfer learning approaches show similar trends. From Stanford→SIRTA, the transfer learning models outperform the local models on both sunny and cloudy days. As shown in the previous experiments on transfer learning to the DEWA dataset, the benefits of local fine-tuning gradually saturate when more training data is available. It also confirms that the knowledge learned from PV power prediction can be transferred to solar irradiance prediction as these two variables are correlated with each other. However, a caveat should be added here. PV power output is affected by the panel temperature (i.e. high temperature can hinder the power generation of PV), which could be learned by the deep learning model from the data, while irradiance measurements are less affected by temperature changes. Therefore, the temperature parameter needs to be taken into account for transfer learning from PV to irradiance. The average temperature in Stanford being below 30 °C in the summer, its impact on solar energy

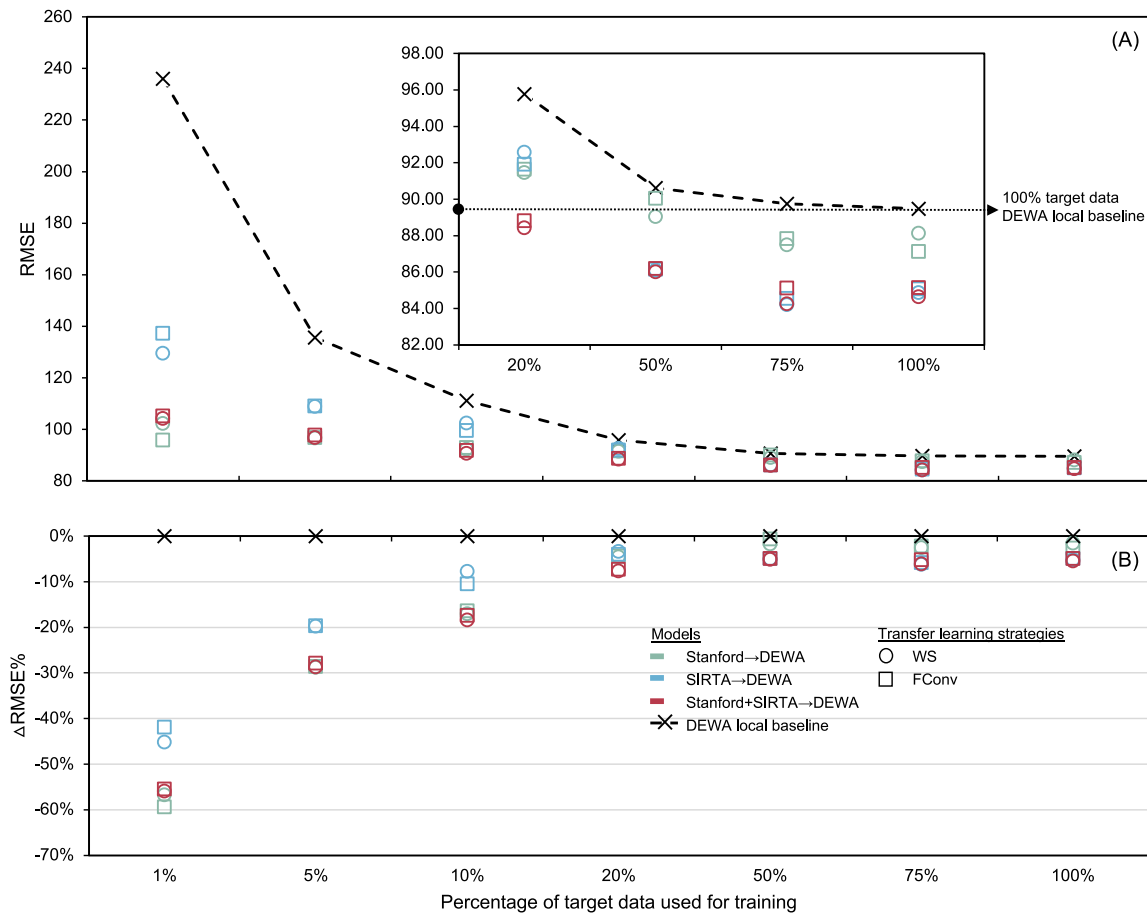


Fig. 6.3. Comparison of different transfer learning strategies and different amounts of data available for the target dataset. The RMSEs are evaluated based on the DEWA test set. Subplot (A) shows the RMSE for models learned using different transfer learning strategies at each level of target data; The insert plot in (A) shows the zoom-in of results from 20% to 100% target data and the dotted line represents the RMSE of the DEWA local baseline model trained using 100% data; The baseline performance is represented by the dashed line in each subplot; Subplot (B) shows the relative performance of the models learned using different transfer learning strategies compared with the DEWA local model baseline for each level of target data. For each target data level, local models correspond to $\Delta RMSE\% = 0\%$, and negative $\Delta RMSE\%$ indicates that the models learned with different transfer learning strategies outperform the corresponding DEWA local model baseline.

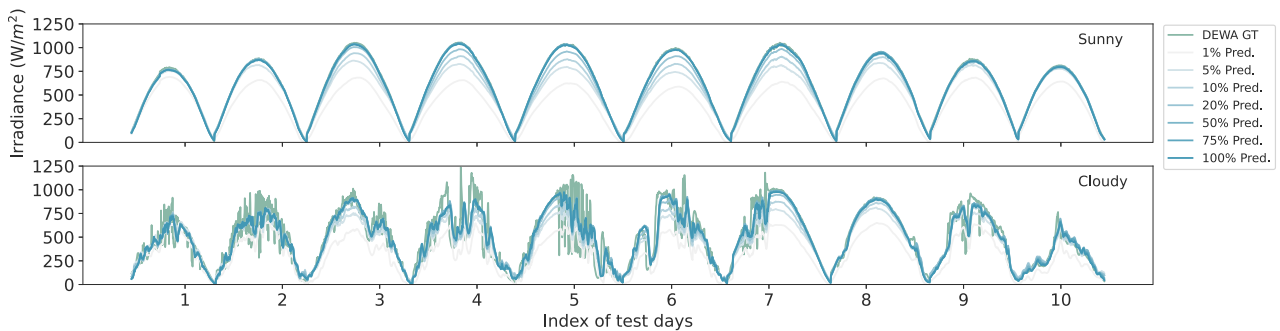


Fig. 6.4. Visualization of predictions (Pred.) from the DEWA local baseline model trained with different amounts of data using the SUNSET architecture versus ground truth (GT) on the DEWA test set. The green curve represents the ground truth. Different shades of blue represent the local models trained with different amounts of data (light blue stands for a small dataset for training, dark blue stands for a large dataset for training). Note that for sunny days, the predictions from model trained with 50% data and above almost overlap with the ground truth. Huge gaps can be observed between model predictions and ground truth if there is less than 20% data for training.

production is not predominant in this study. From SIRTA→Stanford, different behaviors are observed depending on the type of weather: the local models outperform the transfer learning models on sunny days, while the transfer learning models do better on cloudy days. This makes sense as the knowledge transferred from the source model (SIRTA in this case) is mostly trained on cloudy data. The WS or FConv transfer learning strategy might put the model at a sub-optimal starting point and hinder the learning if there is not enough data, while things might improve as more training data is available. In comparison, the Stanford

dataset has a relative more balanced distribution between sunny and cloudy samples. Still, it can be observed that transfer learning from Stanford to SIRTA outperforms the SIRTA local baseline. It thus suggests that for transfer learning, the balance of dataset samples should be paid attention to besides purely the size of the dataset.

Regarding the difference in forecast skill performance between sunny and cloudy days, it is worth mentioning that the magnitude of the RMSE strongly differs between both types of weathers. For instance, the RMSE associated with the SIRTA→Stanford ConvLSTM model in

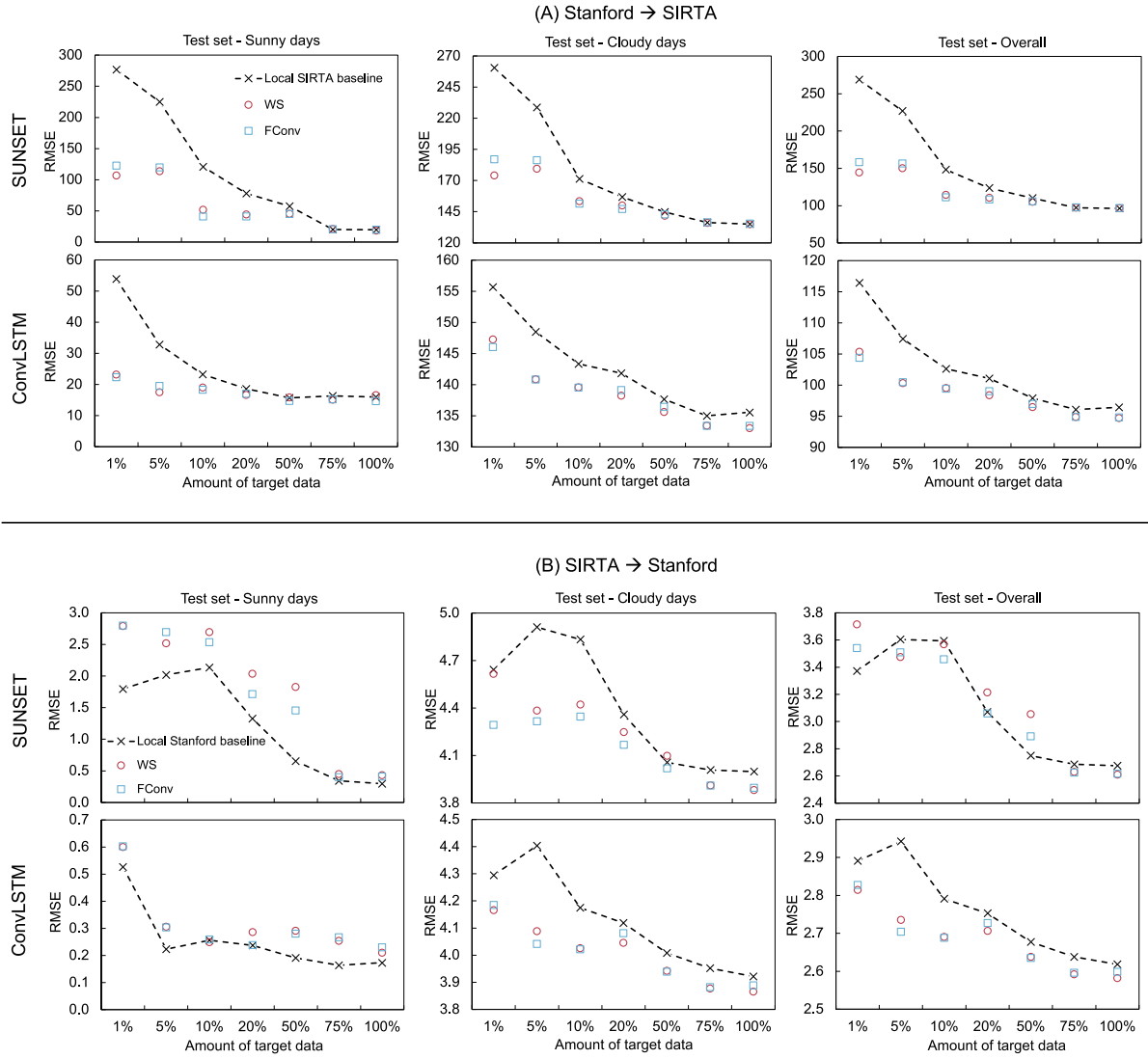


Fig. 6.5. Transfer learning between Stanford dataset and SIRTa dataset for SUNSET and ConvLSTM models with different amounts of target data and different transfer learning strategies. Subplot (A) shows the models pre-trained on the Stanford dataset and transferred to the SIRTa dataset using different strategies; Subplot (B) shows the models pre-trained on the SIRTa dataset and transferred to the Stanford dataset using different strategies.

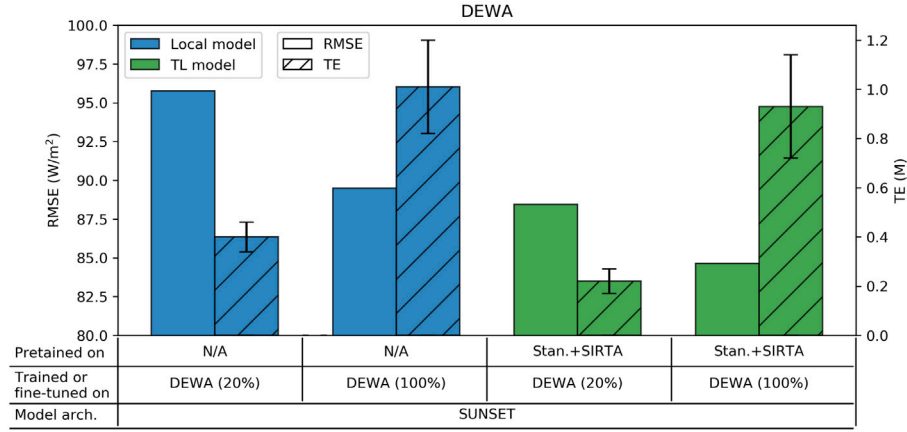
sunny conditions (10 days) remains below 0.3 kW when the amount of training data is above 5%. In comparison, the RMSE in cloudy days (10 days) is about 4 kW. Once squared, the resulting weight of the error corresponding to the 10 sunny days in the total MSE of the test set is below 1%. In other words, doubly the RMSE in sunny days from 0.3 to 0.6 kW only increases the overall test MSE by less than 2%, which is equivalent to a RMSE increase in cloudy days by less than 1%. Consequently, sunny training samples might be marginalized out in a training guided by a MSE loss function.

6.4. Overall evaluation of different training strategies

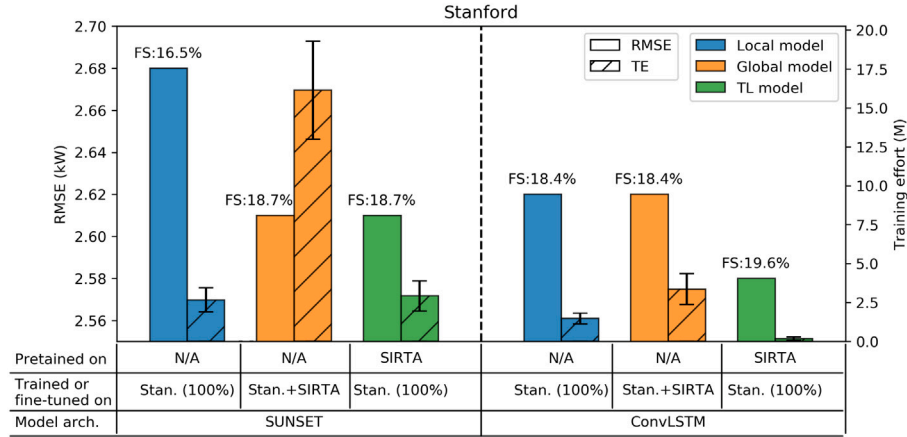
Fig. 6.6 and Table A.1 summarize the performance of local, global and transfer learning models on each one of the three datasets for both SUNSET and ConvLSTM architectures. The performance is evaluated based on both the prediction accuracy (quantified by target test dataset RMSE) as well as the training cost (measured by training effort (TE) as defined in Section 4.2). For the DEWA dataset as the target set, the results show that models trained via transfer learning, perform similarly to the local baseline model with 80% less training data and lower TE. When the whole target training data is available, models trained with transfer learning outperform the local model by 5% with 14% less

training effort. For the Stanford dataset and the SIRTa dataset as the target set, both SUNSET and ConvLSTM architectures are used in the experiments. While the general trends are aligned, the two architectures show distinct behaviors in terms of model convergence, which leads to differences in TE. For both architectures, transfer learning models reach a slightly better prediction accuracy than the local and global models (improvement within 3%) when the whole dataset is utilized. Depending on the model and the direction of the transfer between SIRTa and Stanford datasets, transfer learning strategies seem to impact the TE differently. For the SUNSET model, the TE of transfer learning relative to the baseline local model ranges from -6 to +10%, while for the ConvLSTM model it decreases from -4 to -87%. In particular, the TE is down by a factor of 8 to 9 when ConvLSTM is used to transfer knowledge from SIRTa to Stanford. Although the knowledge acquired by the global model is such that it can be applied to individual locations without compromising its prediction ability (as presented in Table 6.1), its training effort is generally higher than the ones of the local and transfer learning model based on a single dataset due to the increasing dataset size and possibly more training epochs to converge.

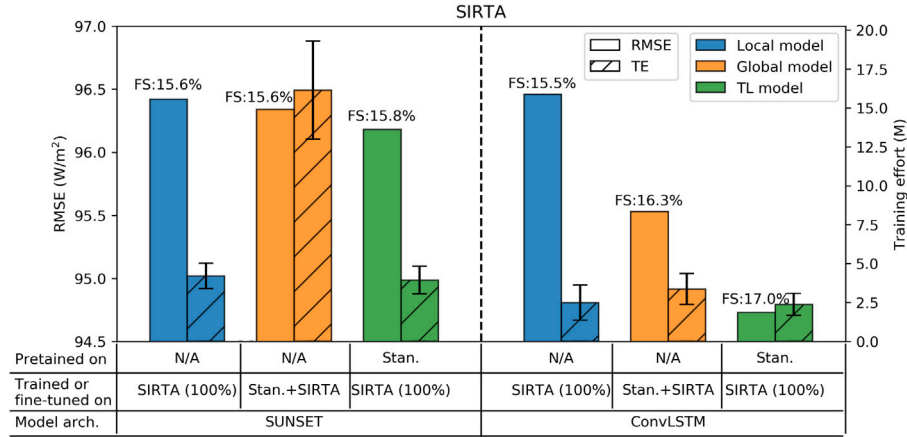
Overall, transfer learning appears to be the best of the three training strategies given its superior performance in terms of both prediction accuracy and computational cost for model training. It is especially



(a) DEWA dataset as target for testing



(b) Stanford dataset as target for testing



(c) SIRTA dataset as target for testing

Fig. 6.6. Performance of different training strategies evaluated by RMSE and training effort on different target datasets. Stan. represents Stanford dataset and TE stands for training effort. The percentage in the brackets next to the dataset names mean the amount of local data used for training or fine-tuning the model. Forecasting skills (%) are annotated on each training strategy for Stanford and SIRTA datasets as target set.

suitable for situations with limited data such as newly operating solar farms. As more and more open-source datasets are made available to the community, this study paves the way to building pre-trained solar forecasting models on a centralized large-scale dataset with numerous and diversified samples (similar to ImageNet [56] for general computer vision applications). This would unlock diverse applications based on transfer learning such as local adaptation of a global expert model

requiring little or even no local training data [69]. As such, transfer learning has the potential to significantly ameliorate the generalization skills of data-driven solar forecasting models based on sky images which currently require a several months of local measurements to train and up to several years of data covering diverse meteorological seasons to reach state-of-the-art performance. The main foreseeable challenges for academia are the collection of open-source sky image datasets from

Table A.1

Summary of the performance of local, global and transfer learning (TL) models for SUNSET and ConvLSTM architectures. The performance is evaluated for each of the dataset based on both the prediction accuracy (measured by target test set RMSE) and the computational cost of training (measured by training effort).

Model architecture	Pre-trained on	Trained or fine-tuned on (% training data)	Model type	Tested on	Test set RMSE (↓)/FS (↑)	Training samples	Training epochs	Training effort
SPM	N/A	N/A	N/A	Stanford Sirta	3.21/0% 114.2/0%	N/A	N/A	N/A
SUNSET	N/A	DEWA (20%)	L	DEWA	95.76	17,056	23.4 ± 3.4	0.40 ± 0.06M
	Stanford+Sirta		TL		88.44	17,056	12.7 ± 2.9	0.22 ± 0.05M
	N/A	DEWA (100%)	L		89.49	85,278	11.8 ± 2.2	1.01 ± 0.19M
	Stanford+Sirta		TL		84.64	85,278	10.9 ± 2.4	0.93 ± 0.21M
SUNSET	N/A	Stanford (100%)	L	Stanford	2.68/16.5%	125,876	21.2 ± 6.1	2.67 ± 0.77M
	Sirta		TL		2.61/18.7%	125,876	23.2 ± 7.7	2.92 ± 0.97M
	N/A	Stanford+Sirta	G		2.61/18.7%	564,048	28.6 ± 5.6	16.13 ± 3.15M
	N/A		L		96.42/15.6%	438,172	9.6 ± 1.8	4.21 ± 0.81M
	Stanford	Sirta (100%)	TL		96.18/15.8%	438,172	9.0 ± 2.0	3.94 ± 0.88M
	N/A		G		96.34/15.6%	564,048	28.6 ± 5.6	16.13 ± 3.15M
ConvLSTM	N/A	Stanford (100%)	L	Stanford	2.62/18.4%	125,876	11.7 ± 2.7	1.47 ± 0.34M
	Sirta		TL		2.58/19.6%	125,876	1.5 ± 0.9	0.19 ± 0.11M
	N/A	Stanford+Sirta	G		2.62/18.4%	564,048	6.0 ± 1.7	3.36 ± 0.99M
	N/A		L		96.46/15.5%	438,172	5.6 ± 2.6	2.49 ± 1.13M
	Stanford	Sirta (100%)	TL		94.73/17.0%	438,172	5.4 ± 1.6	2.38 ± 0.70M
	N/A		G		95.53/16.3%	564,048	6.0 ± 1.7	3.36 ± 0.99M

Notes: (1) For “Model type”, L means local model, TL means transfer learning model, G means global model. The global models are not trained on the target training set. They are only trained on the source datasets and applied to the target test set. (2) The test RMSE is calculated based on the ensemble mean prediction of the ten sub-models from ten-fold cross-validation. (3) For “Training samples”, the meaning is different for different types of models. For local and transfer learning models, the number of training samples used for the target model is counted without counting those used to develop the source model; this is because large-scale pre-trained models will mostly be open-source and accessible, and users do not need to train by themselves. For the global model, the number of training samples for the source model is counted as it is directly applied to the target test set without training a target model. (4) “Training epochs” is the epochs needed until model convergence, which is represented by mean ± std based on ten sub-models from ten-fold cross-validation. (5) Training effort = Training samples × Training epochs, and M stands for a million samples. (6) TL results are reported based on warm-starting strategy.

around the world [34] and the fusion of these heterogeneous datasets into a single training framework, which this study has touched base on.

7. Conclusion

Building solar power modeling algorithms based on multi-location data holds a significant potential for developing more accurate and robust methods. Although this modeling paradigm is common, the data heterogeneity in sky image datasets has not been investigated thoroughly to this day. This study therefore explores the development of deep learning models for solar forecasting based on three heterogeneous datasets collected from around the world. The accuracy of models trained individually based on local datasets (local models) and models trained jointly based on a fusion of multiple heterogeneous datasets (global models) is compared. In addition, the potential of knowledge transfer from pre-trained solar forecasting neural networks to a new dataset of interest (transfer learning models) is examined.

The results indicate that the local models work best when deployed locally. Significant errors are observed regarding the scale of the predicted variable when applied offsite. However, the trend and shape of the PV/irradiance fluctuations are well predicted, which indicates that the feature representations learned by the local models can generally be shared, while the learned regression functions are location dependent. With proper normalization of the prediction target variables, training on a global dataset can improve the generalization of the model, as evidenced by better performances when adapting to individual locations. Nevertheless, one should be aware of the possible increase in training effort required. Besides, we show that the global model is able to differentiate by itself the nuances in the sky images collected from a few locations without the need to inject location information or use location-specific regression heads. Transfer learning tends to accelerate model training and improve solar forecasts, especially when the availability of local data is limited or when the source model is pre-trained on a large and diversified source dataset. For instance, the results show that pre-training solar forecasting algorithms on the combined global Stanford + Sirta dataset and then transferring to the local DEWA dataset

reduces local data requirement by about 80% while decreasing TE five-fold and achieving similar prediction accuracy compared with the local baseline model trained on the whole dataset. Thus, the community is further encouraged to contribute to a large-scale global dataset containing large quantities of diversified sky images for solar forecasting. Open-sourcing pre-trained models built upon such a large-scale dataset can benefit local model development by saving significantly on training effort. Future studies will also investigate the benefit of data augmentation (e.g., image mixing, rotations, vertical/horizontal flips) and scene representation (e.g., polar coordinates, sun-centered sky images) on knowledge transfer in more diverse contexts.

These results provide some answers and insights to the set of questions raised in the introduction:

1. *Can trained deep learning models be applied directly to new sites without local re-training or fine-tuning?*
With the current modeling approach, the generalization of the models beyond their training location is limited. A follow-up work provides some solutions based on the clear-sky index to facilitate transfer learning scenarios between different domains [69].
2. *Should deep learning models be trained locally using the location-specific dataset only or globally via fusion of datasets from different locations?*
When only applied locally, models trained on a local dataset perform similarly to models trained on a global dataset, based on the premise that enough local training data is available. However, global models show significantly higher generalization skills when applied to new sites at the cost of additional TE.
3. *How can global models deal with the dataset heterogeneity, especially the different scales and distributions of prediction targets?*
Data normalization via location-specific time series statistics, multi-regression head approaches and location conditioning strategies appear to facilitate model learning based on heterogeneous datasets.

4. Can knowledge be shared between different locations via transfer learning from pre-trained models?

Pre-trained solar forecasting models fine-tuned on local data train faster and achieve superior performances than models trained on local data only. This indicates the possibility and advantages of sharing knowledge via transfer learning.

Overall, this study lays the first stone in building deep solar forecasting models using multi-location sky image datasets [70], which are faster to train, more accurate and more robust. Based on these results, we hypothesize that future operational algorithms will be developed via so-called foundation models trained on decades of diversified cloud cover observations through various tasks and fine-tuned to specific solar forecasting objectives and sites. To advance the generalization skills of solar forecasting models beyond their training source location, future work will focus on developing transfer learning schemes that require no or limited local data (e.g., one day of sky images and corresponding solar measurements) [69].

CRedit authorship contribution statement

Yuhao Nie: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Quentin Paletta:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Andrea Scott:** Writing – original draft, Validation, Software, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Luis Martin Pomares:** Writing – review & editing, Supervision, Resources, Project administration, Methodology. **Adam Brandt:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The Stanford dataset can be accessed at <https://purl.stanford.edu/dj417rh1007> with detail information documented in <https://github.com/yuhao-nie/stanford-solar-forecasting-dataset>, and the SIRTa dataset is available upon request to https://sirta.ipsl.polytechnique.fr/data_form.html. In addition, we are working on building a large-scale sky image database — SkyImageNet — for solar forecasting and related research. We encourage the readers to check the following GitHub repository for updates: <https://github.com/SkyImageNet/skyimagenet>.

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Appendix. Detailed results for the overall evaluation of the different training strategies

See Table A.1

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